# Quantum Teleportation Simulation

DAY 25 | 30-DAY QUANTUM COMPUTING CHALLENGE | MINI PROJECT

## Project Objective

Simulate the transfer of an arbitrary single-qubit state from Alice to Bob using a shared entangled pair, two classical measurement bits and conditional correction gates. This is state teleportation, not the movement of a particle.

## 1. The Core Idea

Alice holds the input qubit q0. Alice and Bob share entangled qubits q1 and q2. Alice performs a Bell-basis measurement on q0 and q1, then sends Bob two classical bits. Bob applies the corresponding corrections to q2.

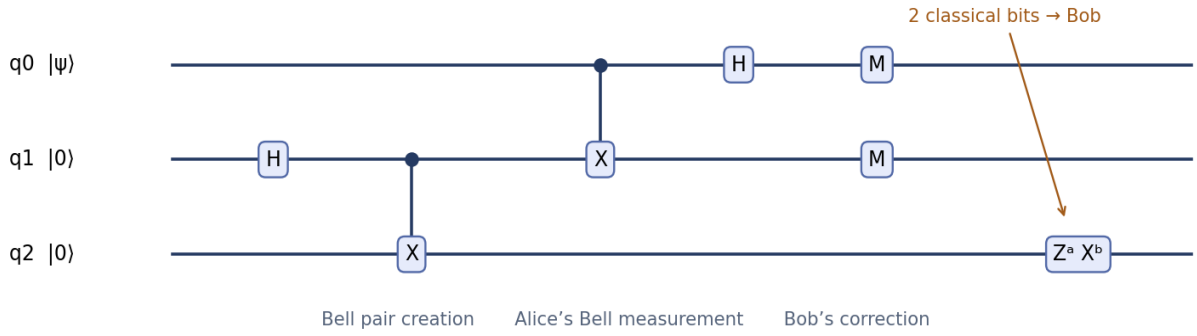



Figure 1. Conceptual three-qubit teleportation workflow (q0 = input, q1 = Alice, q2 = Bob).

## 2. Circuit Execution: Step by Step

• Prepare an arbitrary input state |psi> = cos(theta/2)|0> + exp(i phi) sin(theta/2)|1>. In this experiment theta = pi/3 and phi = pi/5.

• Apply H to q1, followed by CNOT(q1, q2), creating the Bell pair (|00> + |11>)/sqrt(2).

• Apply CNOT(q0, q1), then H(q0). Measure q0 and q1 to obtain two classical bits (a,b).

• Send (a,b) to Bob. He applies X when b = 1 and Z when a = 1. The corrected q2 recovers the original input state.

## 3. Measurement and Correction Table

| Alice bits (q0 q1) | Bob correction | Ideal probability |
|---|---|---|
| 00 | Identity | 25% |
| 01 | X | 25% |
| 10 | Z | 25% |
| 11 | X then Z | 25% |

## 4. What the Experiment Verifies

The notebook computes the full three-qubit statevector, projects each of Alice's four measurement branches, applies Bob's corrections and evaluates state fidelity:

**Fidelity:** F = |<psi | psi_Bob>|^2. In the ideal simulation, F = 1.000 for all four outcomes.

## 5. Measured Results: 1,000 Shots

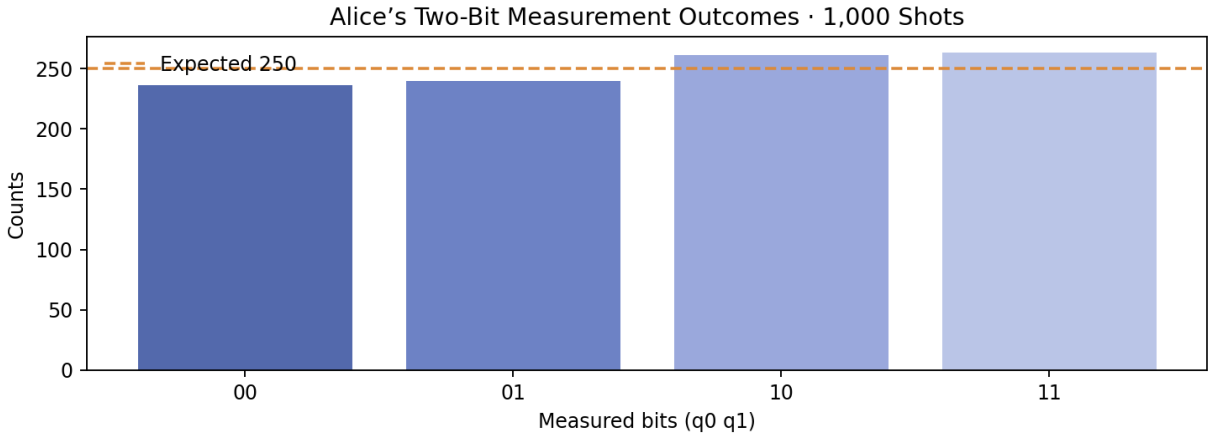



Figure 2. Sampled measurement outcomes; dashed line indicates the expected count of 250 per branch.

## 6. How to Run the Notebook

1. Install Python packages: `pip install numpy matplotlib notebook`.

2. Open `quantum_teleportation_simulation.ipynb` with Jupyter or VS Code.

3. Run all cells to view the input amplitudes, four branch fidelities and measurement histogram.

4. Change theta or phi and repeat. All corrected branch fidelities should remain approximately 1.

5. Optional: install Qiskit to display a measurement-controlled quantum circuit.

## 7. Important Interpretation

The two classical bits are essential: Bob cannot reconstruct the state before receiving them. Alice's original qubit is consumed by measurement, so the protocol does not clone a state or allow faster-than-light messaging.

## Summary: What to Remember

Teleportation uses **one shared Bell pair + two classical bits + Bob's conditional X/Z corrections** to transfer one qubit state. The result is exact only in the ideal noise-free model; physical hardware introduces errors.